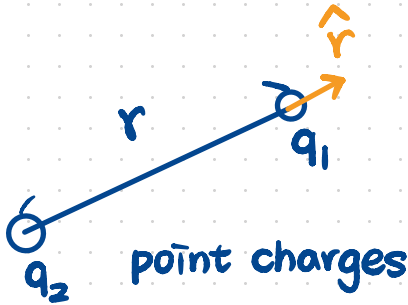


Intended Learning Outcomes (ILOs)

1. Apply Coulomb's Law and the principle of superposition to calculate electrostatic forces.
2. Explain the concept of the electric field and determine the field due to discrete point and continuous charge distributions.
3. Analyze the torque and potential energy of an electric dipole in a uniform electric field.

Ch21 Electric Charge and Electric Field

Coulomb's Law



Electrostatic Force

$$\vec{F}_{2on1} = k \frac{q_1 q_2}{r^2} \hat{r} = -\vec{F}_{1on2}$$

Charges

(1) two types

$q_1 (C)$	$q_2 (C)$	Force (N)
+	+	$\leftarrow \bigcirc \quad \bigcirc \rightarrow$
-	-	
+	-	
-	+	$\bigcirc \rightarrow \quad \leftarrow \bigcirc$

(2) quantized

$$q = ne \quad n \in \mathbb{Z}$$

$$e = 1.602 \times 10^{-19} C \quad \text{elementary charge}$$

(3) Principle of Conservation of Charge:

The algebraic sum of all the electric charges in any closed system is constant.

Note: A closed system can exchange energy but not matter with its surroundings.

(4) How to charge materials

Insulators: electrification by friction

Conductors: charging by induction or contact

Coulomb Constant k

$$k = 8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \equiv \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \quad \text{Permittivity of free space}$$

Inverse - Square Law

$$F = k \frac{q_1 q_2}{r^2}$$

valid for (experiment)
 $r = 10^{-16} \text{ m} \sim 10^8 \text{ m}$

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An α particle (${}^4_2\text{He}^{2+}$) has mass $m = 6.64 \times 10^{-27} \text{ kg}$ and charge $q = +2e = 3.2 \times 10^{-19} \text{ C}$. Compare the magnitude of electric repulsion between two α particles with that of the gravitational attraction between them.

$$F_e = \frac{k q^2}{r^2}$$

$$F_g = \frac{G m^2}{r^2}$$



$$\frac{F_e}{F_g} = \frac{k q^2}{G m^2} = \frac{9.0 \times 10^9 \times (3.2 \times 10^{-19})^2}{6.67 \times 10^{-11} \times (6.64 \times 10^{-27})^2}$$

$$= 3.1 \times 10^{35} *$$

atomic scale ~ 1000 km F_e dominates

universe on large scale F_g dominates

$< 10^{-14}$ m

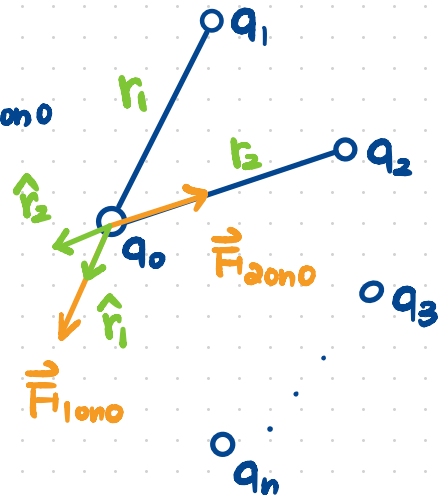
Nuclear Forces dominate

Principle of Superposition

$$\vec{F}_o = \vec{F}_{1o} + \vec{F}_{2o} + \dots + \vec{F}_{no}$$

$$= \sum_{i=1}^n \frac{k q_o q_i}{r_i^2} \hat{r}_i$$

$$= q_o \sum_{i=1}^n \frac{k q_i}{r_i^2} \hat{r}_i$$



$$\vec{F} \equiv q_0 \vec{E}$$

$\downarrow$
 test charge

positive q_0 : $\vec{F}_0$ and $\vec{E}$ are parallel

negative q_0 : $\vec{F}_0$ and $\vec{E}$ are anti-parallel

Electric Field

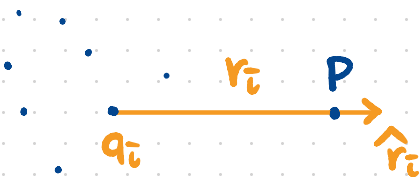
$$\vec{E} \equiv \frac{\vec{F}_0}{q_0}$$

Vector Field

A charged object **sets up** an electric field in the surroundings (even in vacuum)

SI unit : N/C

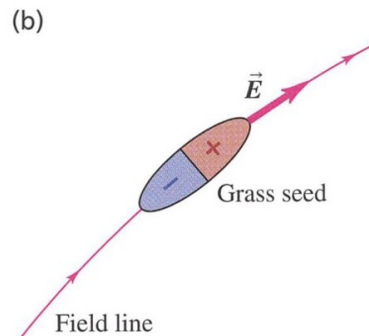
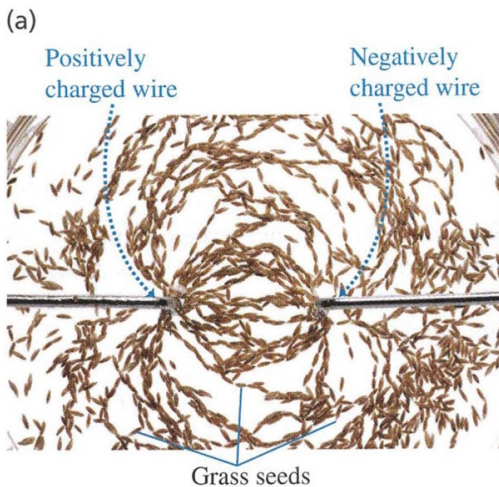
Electric field due to discrete charges



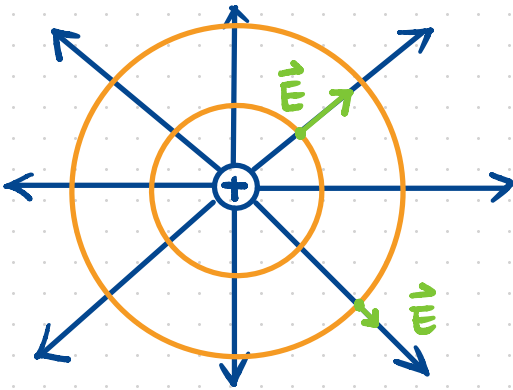
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{r_i^2} \hat{r}_i$$

Electric Field Lines (by Faraday in 19th century)

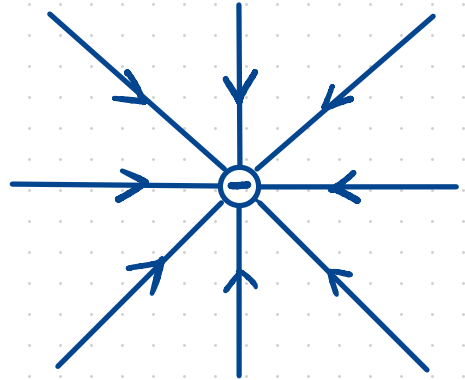
- (1) $\vec{E}$ at a point is **tangential** to the field line through that point.
- (2) At any particular point, the electric field has a **unique direction**.
 $\Rightarrow$ Field lines never intersect.
- (3) $|\vec{E}|$ is proportional to the **density** of field lines in that region.
- (4) $\vec{E}$ extends from **positive charge** to **negative charge**.



Example: Point Charges

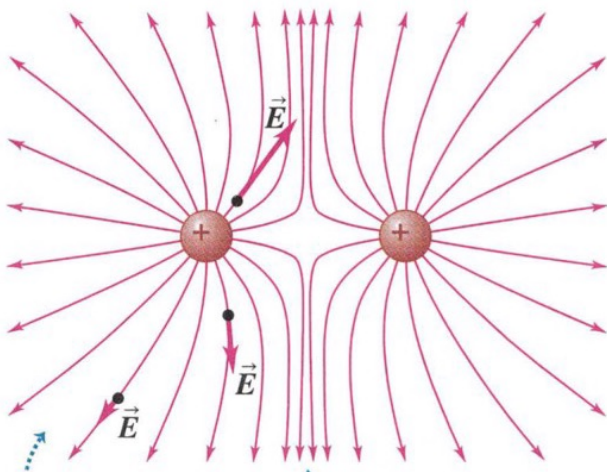


$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

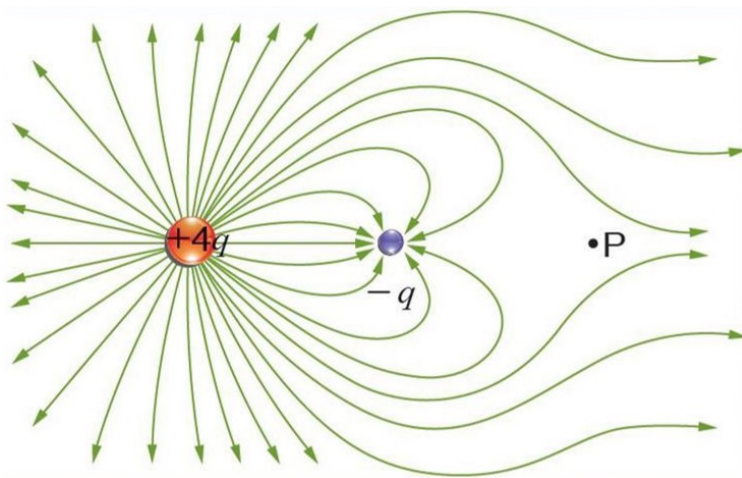


$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{-q}{r^2} \hat{r}$$

Example : Two Point Charges



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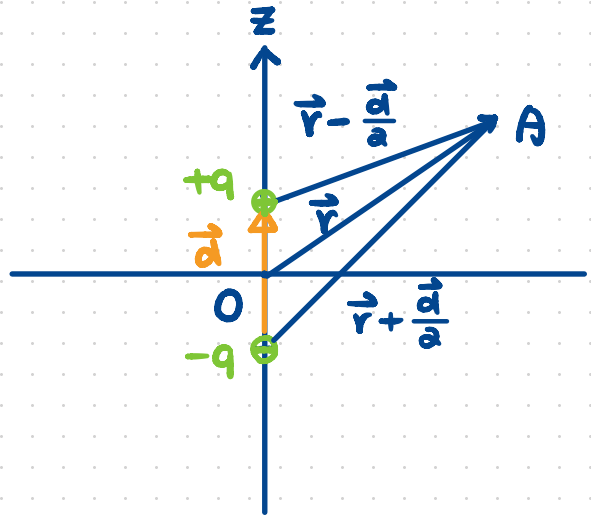
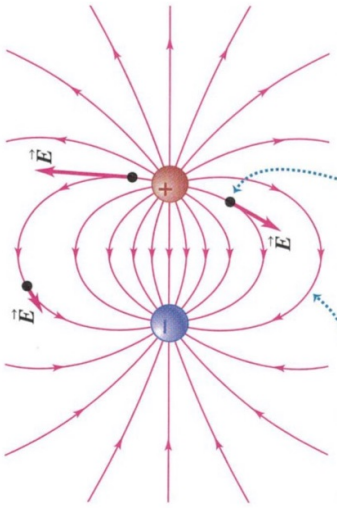


[slideserve.com/erasmus-hutchinson/b-2](https://www.slideserve.com/erasmus-hutchinson/b-2)

Electric - Field Calculations

An Electric Dipole

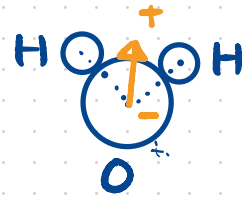
$r \gg d$ Far Field



$$\vec{E}_p = \vec{E}_{p,+q} + \vec{E}_{p,-q}$$

$$= \frac{kq (\vec{r} - \frac{\vec{d}}{2})}{|\vec{r} - \frac{\vec{d}}{2}|^3} - \frac{kq (\vec{r} + \frac{\vec{d}}{2})}{|\vec{r} + \frac{\vec{d}}{2}|^3}$$

e.g. water, heart



$$\stackrel{r \gg d}{=} \frac{kq}{r^3} \left[\left(\vec{r} - \frac{\vec{d}}{2} \right) \left(1 + \frac{3}{2} \frac{\vec{r} \cdot \vec{d}}{r^2} \right) - \left(\vec{r} + \frac{\vec{d}}{2} \right) \left(1 - \frac{3}{2} \frac{\vec{r} \cdot \vec{d}}{r^2} \right) \right]$$

$$= \frac{kq}{r^3} \left[\left(\frac{3\vec{r} \cdot \vec{d}}{r^2} \right) \vec{r} - \vec{d} \right]$$

$$= \frac{kq}{r^3} \left[3(\hat{r} \cdot \vec{d}) \hat{r} - \vec{d} \right]$$

$$= \frac{3(\vec{P} \cdot \hat{r}) \hat{r} - \vec{P}}{4\pi\epsilon_0 r^3} \propto \frac{1}{r^3}$$

Note that

$$\begin{aligned} \left| \vec{r} + \frac{\vec{d}}{2} \right|^{-3} &= \left(r^2 + \vec{r} \cdot \vec{d} + \frac{d^2}{4} \right)^{-\frac{3}{2}} \\ &\stackrel{r \gg d}{=} r^{-3} \left(1 + \frac{\vec{r} \cdot \vec{d}}{r^2} + \frac{d^2}{4r^2} \right)^{-\frac{3}{2}} \\ &\approx r^{-3} \left(1 \pm \frac{3}{2} \frac{\vec{r} \cdot \vec{d}}{r^2} \right) \end{aligned}$$

Electric Dipole Moment

$$\vec{p} \equiv q \vec{d}$$

(1) if $\vec{r} \parallel \vec{p}$ ($\vec{r} = \hat{z}$)

$$z > 0, \hat{r} = \hat{k}, \vec{E}_p = \frac{3p\hat{k} - \vec{p}}{4\pi\epsilon_0|z|^3} = \frac{\vec{p}}{2\pi\epsilon_0|z|^3}$$

$$z < 0, \hat{r} = -\hat{k}, \vec{E}_p = \frac{3\vec{p} \cdot (-\hat{k})(-\hat{k}) - \vec{p}}{4\pi\epsilon_0 r^3} = \frac{\vec{p}}{2\pi\epsilon_0|z|^3}$$

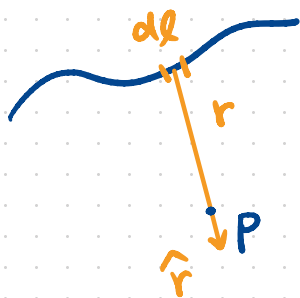
(2) if $\vec{r} \perp \vec{p}$

$$\vec{E}_p = -\frac{\vec{p}}{4\pi\epsilon_0 r^3}$$

Electric field due to continuous charges

$$\vec{E} = \int d\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

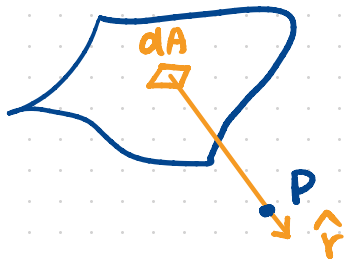
① Line Charge :



$$dq = \lambda d\ell$$

$\lambda(\ell)$: linear charge density (C/m)

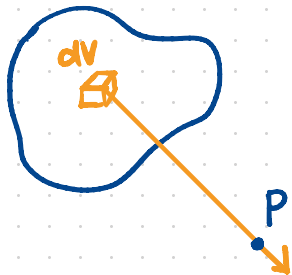
② Surface Charge :



$$dq = \sigma dA$$

$\sigma(x, y, z)$: surface charge density (C/m²)

③ Volume Charge :



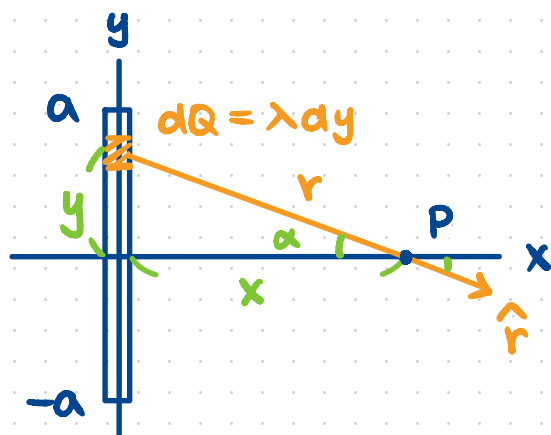
$$dq = \rho dv$$

$\rho(x, y, z)$: volume charge density (C/m³)

Examples of Line Charge :

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Positive charge Q is distributed uniformly along the y -axis between $y = -a$ and $y = +a$. Find the electric field at point P on the x -axis at a distance x from the origin.



$$\lambda = \frac{Q}{2a}$$

$$\vec{E} = E_x \hat{i}$$

$$E_x = \int dE_x$$

$$= \int dE \cos \alpha = \int \frac{k dQ \cos \alpha}{r^2}$$

$$= \frac{\lambda x}{4\pi\epsilon_0} \int_{-a}^a \frac{dy}{(x^2 + y^2)^{3/2}}$$

$$= \frac{\lambda x}{4\pi\epsilon_0} \left[\frac{y}{x^2 \sqrt{x^2 + y^2}} \right]_{-a}^a$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{2a}{x \sqrt{x^2 + a^2}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x \sqrt{x^2 + a^2}}$$

where

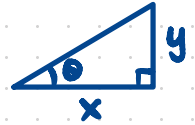
$$dQ = \lambda dy$$

$$\cos \alpha = \frac{x}{r}$$

$$r = \sqrt{x^2 + y^2}$$

Note that $\int \frac{dy}{(x^2 + y^2)^{3/2}}$ $y = x \tan \theta$

$dy = x \sec^2 \theta d\theta$



$$= \int \frac{x \sec^2 \theta d\theta}{x^2 \sec^3 \theta}$$

$$= \frac{1}{x^2} \int \cos \theta d\theta$$

$$= \frac{1}{x^2} \sin \theta + \text{const}$$

$$= \frac{y}{x^2 \sqrt{x^2 + y^2}} + \text{const}$$

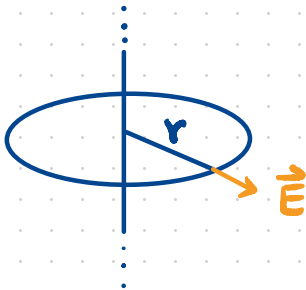
$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x\sqrt{x^2 + a^2}} \hat{i}$$

① Near Field

$$x \gg a \quad \vec{E} \approx \frac{1}{4\pi\epsilon_0} \frac{Q}{x^2} \hat{i}$$

② Infinite Long Line

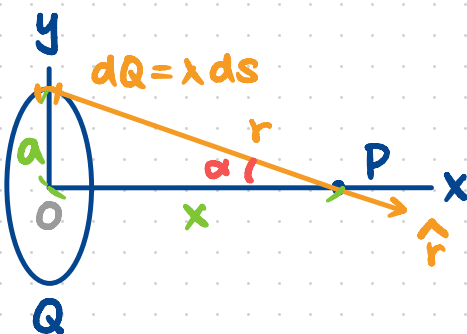
$$a \gg x \quad \vec{E} \approx \frac{1}{4\pi\epsilon_0} \frac{Q}{xa} \hat{i} = \frac{\lambda}{2\pi\epsilon_0 x} \hat{i}$$



$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} \propto \frac{1}{r}$$

It's your turn 😊

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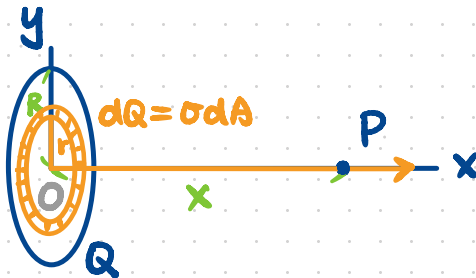
Charge Q is uniformly distributed around a conducting ring of radius a . Find electric field at point P on the ring axis at a distance x from its center.

Examples of Surface Charge:

Example 21.11

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A nonconducting disk of radius R has a uniform positive surface charge density σ . Find the electric field at a point along the axis of the disk a distance x from its center. Assume that x is positive.



$$dQ = \sigma dA = \sigma 2\pi r dr$$
$$\vec{E} = E_x \hat{i}$$

$$E_x = \int dE_x$$

$$= \int \frac{1}{4\pi\epsilon_0} \frac{x dQ}{(x^2 + r^2)^{3/2}}$$

$$= \frac{x}{4\pi\epsilon_0} \int_0^R \frac{\sigma 2\pi r dr}{(x^2 + r^2)^{3/2}}$$

$$= \frac{\sigma x}{4\epsilon_0} \int_{x^2}^{x^2 + R^2} \frac{dt}{t^{3/2}}$$

$$= \frac{\sigma x}{4\epsilon_0} (-2) \left[\frac{1}{\sqrt{x^2 + R^2}} - \frac{1}{x} \right]$$

$$= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{1}{\sqrt{(R/x)^2 + 1}} \right]$$

$$t = x^2 + r^2$$
$$dt = 2r dr$$

Infinite Large Disk

$$R \gg x$$

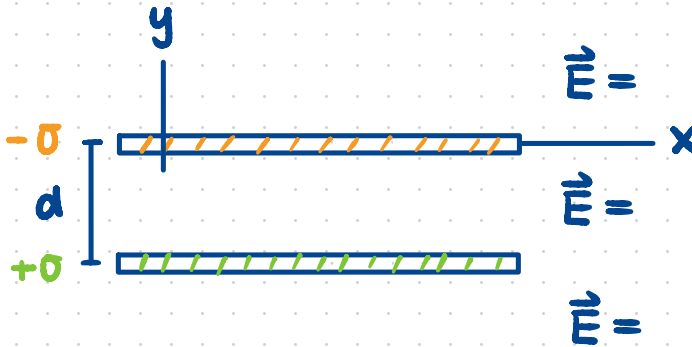
$$\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{i}$$

Independent of
x

Example 21.12

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Two infinite plane sheets with uniform surface charge densities $+\sigma$ and $-\sigma$ are placed parallel to each other with separation d . Find the electric field between the sheets, above the upper sheet, and below the lower sheet.



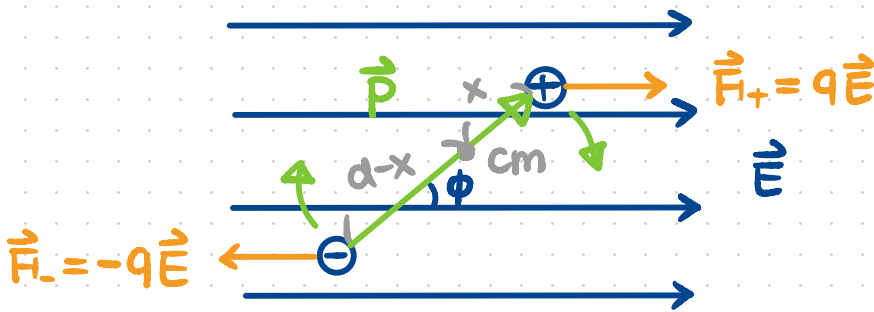
A point charge in an electric field

$$\vec{F} = q \vec{E}$$

positive q : $\vec{F}$ and $\vec{E}$ are parallel

negative q : $\vec{F}$ and $\vec{E}$ are antiparallel

An electric dipole in an uniform electric field



$$\vec{F}_{\text{net}} = \vec{F}_+ + \vec{F}_- = \vec{0}$$

$$|\vec{\tau}| = Fx \sin\phi + F(d-x) \sin\phi$$

$$= Fd \sin\phi$$

$$= qEd \sin\phi$$

$$= pE \sin\phi \text{ (clockwise)}$$

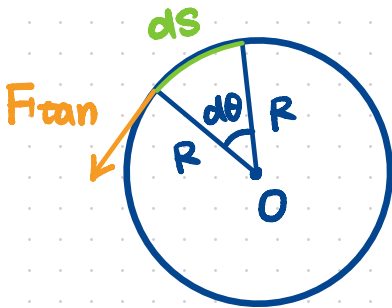
$$\Rightarrow \vec{\tau} = \vec{p} \times \vec{E}$$

Electric force is a conservative force

⇒ define **Electric Potential Energy**

$$\Delta U_e = -W_e$$

Note : Work done by a torque τ_z



$$\begin{aligned} dW &= F_{\text{tan}} ds \\ &= F_{\text{tan}} R d\theta \end{aligned}$$

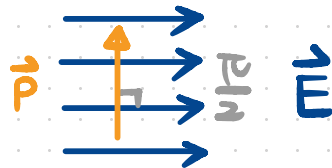
$$\Rightarrow dW = \tau_z d\theta$$

$$W = \int dW = \int_{\theta_1}^{\theta_2} \tau_z d\theta$$

1D Kinematics

$$\Leftrightarrow W = \int F_x dx$$

Set $U_e(\phi = \frac{\pi}{2}) = 0$



$$\Rightarrow U_e(\phi) - \overset{0}{U_e(\phi = \frac{\pi}{2})} = - \int_{\frac{\pi}{2}}^{\phi} \tau_z d\phi'$$

where $\tau_z = -PE \sin \phi'$ (clockwise)

$$\Rightarrow U_e(\phi) = \int_{\frac{\pi}{2}}^{\phi} PE \sin \phi' d\phi'$$

$$= PE (-\cos \phi') \Big|_{\frac{\pi}{2}}^{\phi}$$

$$= -PE \cos \phi$$

$$\Rightarrow \boxed{U_e = -\vec{p} \cdot \vec{E}}$$

Minimum $U_e = -PE$ when $\phi = 0$ 

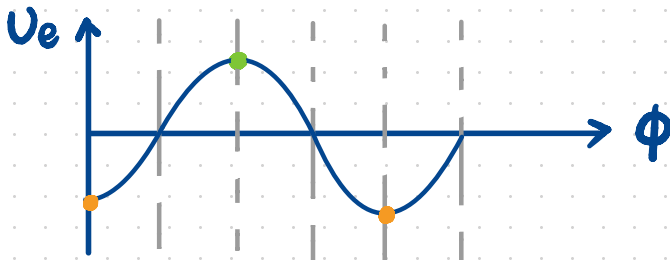
Maximum $U_e = PE$ when $\phi = \pi$ 

Both of them are equilibrium position.

$$\tau_z = - \frac{dU_e}{d\phi} = -PE \sin \phi$$

$$\tau_z = 0 \quad \text{when } \phi = 0 \text{ \& } \pi$$

unstable equilibrium



stable equilibrium

